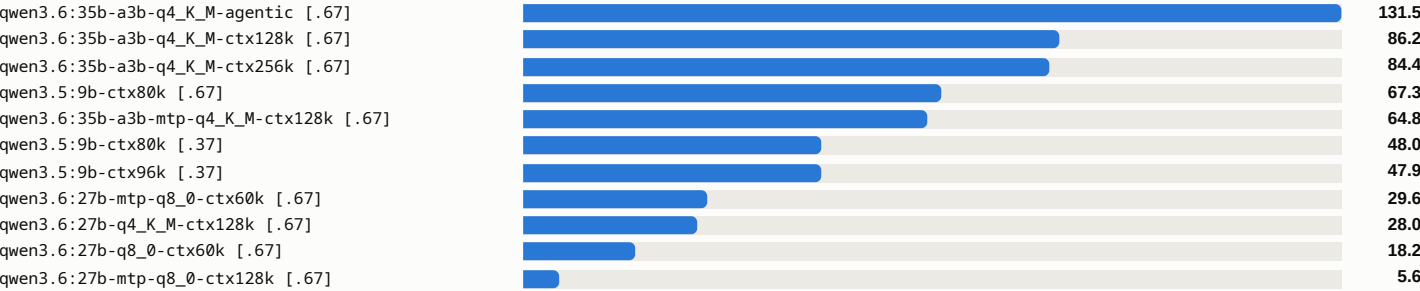


Local models for agentic coding — servers 192.168.100.67 and .37

.67 — 36.1 GB usable VRAM (measured, two GPUs), Ollama 0.32.5 · .37 — 12.2 GB, Ollama 0.30.6 · generated 2026-08-06 · all figures measured on these hosts

Use qwen3.6:35b-a3b-q4_K_M with num_ctx 262144, temperature 0, presence_penalty 0. Measured 131.5 tok/s, 33.08 GB fully resident, the full 256k context in VRAM, all ten agentic gates passed, and a fact recalled from 146957 prompt tokens. That is 7.2× the qwen3.6:27b-q8_0 originally installed on this box — same hardware, three configuration decisions: pick the MoE (3.1× over dense q4 of similar size), clear presence_penalty (1.53×), and bake a num_ctx that fits entirely in VRAM (avoids a 5.3× loss).

GENERATION THROUGHPUT — PROFILE L, THINKING DISABLED (TOK/S)



FOOTPRINT, SPEED AND CONTEXT CEILING

MODEL	SIZE GB	VRAM GB	PLACEMENT	TOK/S S	TOK/S L	MAX CTX IN VRAM	THINKS BY DEFAULT
qwen3.6:35b-a3b-q4_K_M-ctx128k [.67]	28.25	28.25	full	80.0	86.2	262144	yes
qwen3.6:27b-q4_K_M-ctx128k [.67]	30.43	30.43	full	27.8	28.0	131072	yes
qwen3.6:27b-q8_0-ctx60k [.67]	32.87	32.87	full	18.1	18.2	65536	yes
qwen3.6:27b-mtp-q8_0-ctx60k [.67]	31.82	31.82	full	48.8	29.6	32768	yes
qwen3.6:27b-mtp-q8_0-ctx128k [.67]	36.65	32.07	split	11.1	5.6	32768	yes
qwen3.5:9b-ctx80k [.67]	8.79	8.79	full	65.1	67.3	262144	yes
qwen3.6:35b-a3b-mtp-q4_K_M-ctx128k [.67]	26.14	26.14	full	51.4	64.8	262144	yes
qwen3.6:35b-a3b-q4_K_M-ctx256k [.67]	33.08	33.08	full	82.2	84.4	262144	yes
qwen3.6:35b-a3b-q4_K_M-agentic [.67]	33.08	33.08	full	127.2	131.5	262144	yes
qwen3.5:9b-ctx80k [.37]	8.01	8.01	full	47.7	48.0	131072	yes
qwen3.5:9b-ctx96k [.37]	8.56	8.56	full	47.5	47.9	131072	yes

AGENTIC CAPABILITY GATES

MODEL	SINGLE TOOL	PICK RIGHT TOOL	MULTI-TURN	PARALLEL CALLS	NESTED SCHEMA	TOOLS @ LONG CTX	DEEPEST RECALL
qwen3.6:35b-a3b-q4_K_M-ctx128k [.67]	pass	pass	pass	pass	pass	pass	72k tok
qwen3.6:27b-q4_K_M-ctx128k [.67]	pass	pass	pass	pass	pass	pass	72k tok
qwen3.6:27b-q8_0-ctx60k [.67]	pass	pass	pass	pass	pass	pass	18k tok
qwen3.6:27b-mtp-q8_0-ctx60k [.67]	pass	pass	pass	pass	pass	pass	18k tok
qwen3.6:27b-mtp-q8_0-ctx128k [.67]	pass	pass	pass	pass	pass	pass	72k tok
qwen3.5:9b-ctx80k [.67]	pass	pass	pass	pass	pass	pass	72k tok
qwen3.6:35b-a3b-mtp-q4_K_M-ctx128k [.67]	pass	pass	pass	pass	pass	pass	72k tok
qwen3.6:35b-a3b-q4_K_M-ctx256k [.67]	pass	pass	pass	pass	pass	pass	147k tok
qwen3.6:35b-a3b-q4_K_M-agentic [.67]	pass	pass	pass	pass	pass	pass	147k tok
qwen3.5:9b-ctx80k [.37]	pass	pass	pass	pass	pass	pass	72k tok
qwen3.5:9b-ctx96k [.37]	pass	pass	pass	pass	fail	pass	72k tok

HOW TO READ THE GATES

- **nested schema** — enums plus an array of objects, like a real patch tool. **pick right tool** — four offered; must choose search_code.
- **tools @ long ctx** — tool call with ~53k tokens already in the window, run 3×. This is where models silently stop emitting tool calls.
- **deepest recall** — deepest prompt, in measured tokens, from which a fact buried mid-document was still retrieved. Lower figures are window limits, not model limits: the deeper needle overflowed num_ctx and was discarded before the model saw it.
- All gates ran with thinking disabled and a 4000-token budget. Under a squeezed budget these same models fail in ways that look like incapability — no call, wrong schema — so both must be controlled for.

FINDINGS

- **presence_penalty 1.5 — a vendor default — costs 35% of throughput.** Isolated at identical weights/ctx/VRAM: 84.4 → 129.5 tok/s at 0. temperature 0 adds no speed.
- **MTP reverses sign by architecture.** +170% on the bandwidth-bound dense q8 (18.1 → 48.8 S) but –25% on the MoE, which was never bandwidth-starved. Not a generic optimisation.
- **A 12.5% VRAM spill costs 5.3×.** Same weights at ctx60k vs ctx128k: 29.6 → 5.6 tok/s. Loads fine, passes every gate, silently five times slower.
- **Bare model tags cap at 16384 tokens** through /v1/messages, which has no num_ctx knob — and tool calling stops entirely past it, with no error. Only point an agent at a ctx-baked variant.
- **Overflowing num_ctx discards half the window** (num_ctx/2+2; four models, three window sizes) — but only on Ollama 0.32.5. On 0.30.6 the window fills normally, so this is a regression in the newer server.
- **temperature 1 makes small-model tool calls a coin flip.** The 9b passes the nested-schema gate 4–5/8 at defaults, 8/8 at temperature 0 — so single-trial gate results below measure luck as much as capability.
- **The new server is 1.4× the old one,** not the ~2× the raw quoted figures implied (identical model on both hosts). The rest was model and configuration choice.

Profile S = Sieve-of-Eratosthenes prompt capped at 300 tokens (comparable to the v0 history). Profile L = "Write exactly 1000 tokens about GPUs", uncapped (comparable to the figures quoted by Alex). The two are never averaged: longer generations amortise warm-up and report higher tok/s on identical hardware. Context ceiling = largest num_ctx whose weights+KV still sit entirely in VRAM. Recall depth is the deepest prompt, in tokens reported by the server, from which the buried fact was retrieved; prose measures 1.65 tokens/word here. Overflowing num_ctx discards down to num_ctx/2 with no error, so size num_ctx at twice the context you intend to use. GPU temperature and utilisation are not shown because the Ollama HTTP API does not expose them and SSH to the host is refused.